For two integers $a, b \in \mathbb{Z}$, we write $a \mid b$ for "a divides b" (that is, $\exists q \in \mathbb{Z} : aq = b$). We write $\mathbb{Z}^+$ for the set of positive integers $\{1, 2, 3, \ldots\}$. For $M \in \mathbb{Z}^+$, we define the set $\mathbb{Z}_M \coloneqq \{1, 2, \ldots M-1\}$. Recall the elementary rule that we will use throughout: if $a, b, a', b', M \in \mathbb{Z}^+$ such that $a \equiv a' \pmod{M}$ and $b \equiv b' \pmod{M}$, then $ab \equiv a'b' \pmod{M}$.

1 Coprimality and Greatest Common Divisor

Definition 1.1 (Greatest Common Divisor)**.** *Let a and b be integers, not both zero. The greatest common divisor (GCD) of a and b, denoted $\gcd(a, b)$, is the largest integer d such that $d \mid a$ and $d \mid b$.*

Definition 1.2 (Coprime)**.** *Two integers a and b are said to be coprime (or relatively prime) if $\gcd(a, b) = 1$.*

Examples. Consider the integers 15 and 28. The divisors of 15 are $\{1, 3, 5, 15\}$ and the divisors of 28 are $\{1, 2, 4, 7, 14, 28\}$. The only common divisor is 1, thus $\gcd(15, 28) = 1$, making them coprime. Notice that, $\gcd(15, 25) = 5$, so they are not coprime.

2 Bézout's Identity

Theorem 2.1 (Bézout's Identity)**.** *Let a and b be integers, not both zero. There exist integers x and y such that*

$$ax + by = \gcd(a, b)$$

The integers x and y are known as Bézout's coefficients. In the previous section, we saw that $\gcd(15, 28) = 1$. By Bézout's identity, there exist integers x, y such that $15x + 28y = 1$. Check that $x = 15$ and $y = -8$ satisfy this equation. You may refer to wikipedia for a proof. Do other solutions also exist? How can we algorithmically find these integers?

3 The Extended Euclidean Algorithm

The Extended Euclidean Algorithm (EEA) can be used to compute the GCD of two numbers, as well as the Bézout's coefficients. Recall the Euclidean algorithm. There we repeatedly solve some linear equations over integers until one of the coefficients reduce to zero. EEA is basically the Euclidean algorithm that does some additional book to keep track of the coefficients we get during these process. Here is the algorithm.

Algorithm 1 Extended Euclidean Algorithm

Input: Integers a and b, with $a \geq b \geq 0$

Output: Integers g, x, y such that $g = \gcd(a, b)$ and $ax + by = g$

$r_0 \leftarrow a, r_1 \leftarrow b$
$s_0 \leftarrow 1, s_1 \leftarrow 0$
$t_0 \leftarrow 0, t_1 \leftarrow 1$
$i \leftarrow 1$
while $r_i > 0$ **do**
    $q_i \leftarrow \lfloor r_{i-1}/r_i \rfloor$
    $r_{i+1} \leftarrow r_{i-1} - q_i r_i$
    $s_{i+1} \leftarrow s_{i-1} - q_i s_i$
    $t_{i+1} \leftarrow t_{i-1} - q_i t_i$
    $i \leftarrow i + 1$
end while
return $r_{i-1}, s_{i-1}, t_{i-1}$

The while loop in algorithm 1 is executed $\mathcal{O}(\log b)$ times, and the total time complexity is dominated by integer multiplication, coming out to $\mathcal{O}\left(\log^2 a\right)$.

As an exercise, find $\gcd(28, 15)$ and the corresponding Bézout coefficients. We should get that $x = 7$ and $y = -13$. Recall that in the previous example in section 2, we used $x = 15$ and $y = -8$. Thus, these coefficients are not unique. In fact, notice that for the equation $ax + by = c$, and any integer k, one can add and subtract $k \cdot b$ and $k \cdot a$ from x and y respectively to get other integer solutions.

Remark. *Notice that the equation $ax + by = c$ over integers has integer solutions iff the $\gcd(a, b) \mid c$.*

To see this, suppose $d = \gcd(a, b) \nmid c$. Suppose $d \neq 1$ otherwise the the statement is vacuously true. Since d divides both a and b, d divides the left hand side on the equation. Dividing both sides of the equation by d says that an integer is equal to a non-integer rational number, which is a contradiction.

4 The Multiplicative Group $\mathbb{Z}_M^*$

In this course we will restrict ourselves to finite, commutative groups.

Definition 4.1 ((Commutative) Group)**.** *A (Commutative) group $(G, \cdot)$ is a set G, along with a binary operation $\cdot$ that satisfies the following properties:*

1. ***Closure:*** $\forall a, b \in G,\ a \cdot b \in G$.
2. ***Associativity:*** $\forall a, b, c \in G,\ (a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. ***Identity:*** $\exists e \in G : \forall a \in G,\ e \cdot a = a \cdot e = a$. *We refer to e as the group identity.*
4. ***Invertability:*** $\forall a \in G, \exists b \in G : a \cdot b = b \cdot a = e$. *We refer to b as the inverse of a.*

We will be especially worried about the following group.

Definition 4.2 (The Multiplicative Group of Integers Modulo M)**.** *Let M be a positive integer. We define the set $\mathbb{Z}_M^*$ as the set of non-zero integers in $\mathbb{Z}_M$ that are coprime to M:*

$$\mathbb{Z}_M^* = \{a \in \mathbb{Z}_M \mid \gcd(a, M) = 1\}.$$

Notice that in general $\mathbb{Z}_M^* \neq \mathbb{Z}_M$. For integers in $\mathbb{Z}_M^*$, consider the binary operation "multiplication modulo M". (That is, $a \cdot b = ab \mod M$.)

Lemma 4.3. *$\mathbb{Z}_M^*$ forms a group under the binary operation multiplication modulo M.*

Proof Sketch. Check that closure holds because if $\gcd(a, M) = 1$ and $\gcd(b, M) = 1$, then $\gcd(ab, M) = 1$. Associativity follows from standard integer multiplication. The identity element is 1, since $\forall a \in \mathbb{Z}_M^*, \gcd(1, a) = 1$.

To prove invertibility, consider any $a \in \mathbb{Z}_M^*$. Notice that $\gcd(a, M) = 1$, and apply Bézout's identity. There exist integers x, y such that

$$ax + My = 1. \tag{4.1}$$

Taking this equation modulo M yields $ax \equiv 1 \pmod{M}$. The integer x mod M is therefore the multiplicative inverse of a in $\mathbb{Z}_M^*$ (check that $x \mod M$ also satisfies eq. (4.1)). ☐

Example. Let $M = 15$. Then, $\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$. Notice that $2 \times 8 = 16 \equiv 1 \pmod{15}$, meaning 2 and 8 are inverses of each other in this group.

5 Euler's Totient Function

Definition 5.1 (Euler's Totient Function)**.** *For a positive integer M, Euler's totient function, denoted $\varphi(M)$, is the number of integers in the range $1 \leq k \leq M$ that are coprime to M. Equivalently, it is the size of the group $\mathbb{Z}_M^*$, so $\varphi(M) = |\mathbb{Z}_M^*|$.*

Theorem 5.2. *If $M \in \mathbb{Z}^+$ is prime, then $\varphi(M) = M - 1$.*

Proof. If M is prime, its only positive divisors are 1 and M. Therefore, $\forall k \in \mathbb{Z}_M, \gcd(k, M) = 1$. $|\mathbb{Z}_M| = M - 1$, therefore, $\varphi(M) = M - 1$. ☐

Theorem 5.3. *If p and q are prime numbers and $M = pq$, then $\varphi(M) = (p - 1)(q - 1)$.*

Proof. We want to find the number of integers in $\mathbb{Z}_M$ that are coprime to pq. We will subtract from pq the number of elements that share a factor with pq. Since p and q are prime numbers, the integers that share a factor with pq are exactly the multiples of p and the multiples of q. (To see this, notice that if there were any other such integers, they would share a factor with p, or with q. This would contradict with their primality.)

The multiples of p in that range are exactly $p, 2p, 3p, \ldots, qp$. There are exactly q such multiples. Similarly, the multiples of q are $q, 2q, 3q, \ldots, pq$. There are exactly p such multiples. But the integer pq is counted twice. Therefore, the number of integers not coprime to pq is $p + q - 1$. We get that

$$\varphi(pq) = pq - (p + q - 1) = pq - p - q + 1 = (p - 1)(q - 1).$$

☐

6 Euler's Totient Theorem

Theorem 6.1 (Euler's Theorem)**.** *If $a, M \in \mathbb{Z}^+$ are coprime, then*

$$a^{\varphi(M)} \equiv 1 \pmod{M}.$$

Proof. Let $\mathbb{Z}_M^* = \{x_1, x_2, \dots, x_k\}$, where $k = \varphi(M)$. Since a is coprime to M, $a \in \mathbb{Z}_M^*$. Consider

$$a \cdot \mathbb{Z}_M^* \coloneqq \{ax_1, ax_2, \dots, ax_k\},$$

the set we get by multiplying every element in the group by a. Because $\mathbb{Z}_M^*$ is a group, multiplication by a simply permutes the elements of the group. (To see this, notice that beacuse $a \in \mathbb{Z}_M^*$, it has an inverse, and thus we have that $\forall x_i, x_j \in \mathbb{Z}_M^*,\ ax_i \equiv ax_j \pmod{M}$ if and only if $x_i = x_j$.) Therefore, the product of all elements in the original set is congruent modulo M to the product of all elements in the new set:

$$\prod_{i=1}^{k} x_i \equiv \prod_{i=1}^{k} (ax_i) \pmod{M}$$

$$\prod_{i=1}^{k} x_i \equiv a^k \prod_{i=1}^{k} x_i \pmod{M}$$

Since each x_i is coprime to M, their product is also coprime to M. This means the product has a multiplicative inverse modulo M. Multiplying both sides by this inverse and substituting k completes the result,

$$1 \equiv a^{\varphi(M)} \pmod{M}.$$

□

7 Order and Generators

Definition 7.1 (Order of an Element). *Let $a \in \mathbb{Z}_M^*$. The order of a, denoted* $\mathrm{ord}_M(a)$*, is defined as the smallest positive integer r such that $a^r \equiv 1 \pmod{M}$.*

Lemma 7.2 (Order Divides Totient). *For any $a \in \mathbb{Z}_M^*$, the order* $\mathrm{ord}_M(a)$ *divides $\varphi(M)$.*

Proof. Let $r = \mathrm{ord}_M(a)$. By theorem 6.1, we know $a^{\varphi(M)} \equiv 1 \pmod{M}$. Using the division algorithm, we can write $\varphi(M) = q \cdot r + s$, where q is the quotient and $0 \le s < r$ is the remainder. Substituting this into the exponent yields

$$1 \equiv a^{\varphi(M)} \equiv a^{q \cdot r + s} \equiv (a^r)^q \cdot a^s \pmod{M}.$$

Since $a^r \equiv 1 \pmod{M}$, this simplifies to

$$1 \equiv (1)^q \cdot a^s \equiv a^s \pmod{M}.$$

We have $a^s \equiv 1 \pmod{M}$ with $0 \le s < r$. However, r was defined as the *smallest* positive integer that yields 1. Therefore, s cannot be strictly positive; it must be that $s = 0$. This implies $\varphi(M) = q \cdot r$, meaning r divides $\varphi(M)$. □

Recall that for a prime number p we have that $\varphi(p) = p - 1$. Therefore we have the following corollary.

Corollary 7.3. *Let p be a prime number. Then,* $\mathrm{ord}_p(a)$ *divides $p - 1$.*

Definition 7.4 (Cyclic Group and Generator). *A commutative group $(G, \cdot)$ is said to be* cyclic *if $\exists g \in G$ such that $\forall a \in G$, there exists an integer k such that $a = g^k$.*

Such an element g is called a generator *of the group G, and we write $G = \langle g \rangle$ indicating that G is* generated *by g.*

In the context of the multiplicative group $\mathbb{Z}_M^*$, a generator is often called a *primitive root modulo M*. As an example, consider the multiplicative group $\mathbb{Z}_5^* = \{1, 2, 3, 4\}$. Let us examine the powers of the element 2 modulo 5:

- $2^1 \equiv 2 \pmod 5$
- $2^2 \equiv 4 \pmod 5$
- $2^3 = 8 \equiv 3 \pmod 5$
- $2^4 = 16 \equiv 1 \pmod 5$

Since the powers of 2 generate every element in the set $\{1, 2, 3, 4\}$, the element 2 is a generator. Therefore, $\mathbb{Z}_5^*$ is a cyclic group, and we can write $\mathbb{Z}_5^* = \langle 2 \rangle$. In the remainder of this section, we will prove theorem 7.7, which says that if p is a prime number, then $\mathbb{Z}_p^*$ is a cyclic group. We will first prove some auxiliary results.

Lemma 7.5. *For any integer $M \geq 1$, the sum of the totient function over all positive divisors d of M is equal to M:*

$$\sum_{d \in \mathbb{Z}_M : d | M} \varphi(d) = M$$

Proof. Consider the M *distinct* rational numbers $\frac{1}{M}, \frac{2}{M}, \ldots, \frac{M}{M}$. Let $\frac{k_i}{d_i}$ denote the i^{th} such fraction in the irreducible form. Notice that the denominator of each reduced fraction must be some divisor d_i of M.

For any specific divisor d, the number of fractions that have d as their denominator when reduced is exactly the number of numerators $k \leq d$ such that $\gcd(k, d) = 1$ (since otherwise $\frac{k}{d}$ could be reduced further.) By definition, there are exactly $\varphi(d)$ such numerators. Since every original fraction reduces to a unique fraction with a denominator d dividing M, we get that $\sum_{d|M} \varphi(d) = M$. $\square$

Let p be a prime, and let d be a divisor of $p - 1$. Let $N_p(d)$ denote the number of elements in $\mathbb{Z}_p^*$ that have an order exactly equal to d. We will first prove that $N_p(d) \in \{ 0, \varphi(d) \}$.

Lemma 7.6. *If $N_p(d) > 0$, then $N_p(d) = \varphi(d)$.*

Proof. Assume $N_p(d) > 0$, meaning there exists at least one element $a \in \mathbb{Z}_p^*$ with $\text{ord}_p(a) = d$. Consider the sequence of elements $1, a, a^2, \ldots, a^{d-1}$. Since a has order d, these d elements are all distinct modulo p. Furthermore, for any $k \in \{0, 1, \ldots, d-1\}$, $(a^k)^d = (a^d)^k \equiv 1^k \equiv 1 \pmod p$. Thus, these d elements are all roots of the polynomial equation $x^d - 1 \equiv 0 \pmod p$. Since a polynomial of degree d modulo a prime can have at most d roots, these are *all* the roots of that equation.

Any element $b \in \mathbb{Z}_p^*$ of order d must satisfy $b^d \equiv 1 \pmod p$, so b must be one of the elements a^k. The order of an element a^k is given by $\frac{d}{\gcd(k,d)}$. For a^k to have an order of exactly d, it must be that $\gcd(k, d) = 1$. The number of such integers k between 0 and $d - 1$ coprime to d is exactly $\varphi(d)$. Therefore, the number of elements of order d is exactly $\varphi(d)$. $\square$

Theorem 7.7 (Existence of Generators)**.** *For any prime p, the multiplicative group $\mathbb{Z}_p^*$ is cyclic.*

Proof. We prove that there exists at least one element $a \in \mathbb{Z}_p^*$ such that $\text{ord}_p(a) = p - 1$, making a a generator of the group. From theorem 7.6, we know that for any divisor d of $p - 1$, $N_p(d)$ is either 0 or $\varphi(d)$. Thus, $N_p(d) \leq \varphi(d)$ for all d. Summing this inequality over all divisors d of $p - 1$, we get:

$$\sum_{d \in \mathbb{Z}_p : d | p-1} N_p(d) \leq \sum_{d \in \mathbb{Z}_p : d | p-1} \varphi(d).$$

We have the following claim.

Claim 7.8. *For any prime p,*

$$\sum_{d \in \mathbb{Z}_p : d | (p-1)} N_p(d) = p - 1. \tag{7.1}$$

To prove the claim, recall that by theorem 7.2, the order of every element in $\mathbb{Z}_p^*$ must divide $\varphi(p) = p-1$. Every element in $\mathbb{Z}_p^*$ has exactly one order, and there are exactly $p-1$ elements. The result follows.

Theorem 7.8 implies that the left side equals $p-1$. From theorem 7.5, the right side also equals $p-1$. Since the sum of $N_p(d)$ exactly equals the sum of $\varphi(d)$, and no individual term $N_p(d)$ can exceed $\varphi(d)$, it must be strictly true that $N_p(d) = \varphi(d)$ for every divisor d. In particular, for $d = p-1$, we have $N_p(p-1) = \varphi(p-1)$. Since $p \geq 2$, $\varphi(p-1) \geq 1$. Therefore, there exists at least one element of order $p-1$. Such an element generates the entire group $\mathbb{Z}_p^*$. $\square$